Mariam Farzana

TV Engineer | Broadcast and Electrical Engineering | Dhaka, Bangladesh

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Professional Summary

Electrical and electronic engineer (MSc, University of Dhaka) with 6 years as a TV Engineer at Bangladesh Television (BTV), covering digital broadcasting infrastructure, electrical power systems and quality assurance of digital services. Works effectively with multidisciplinary engineering, QA and software teams. Additional skills in full-stack web development.

Technical Skills

- Broadcasting: digital terrestrial television (DTT), analogue-to-digital transmitters, UHF antenna systems
- Electrical Power: substations, generators, UPS, AVR
- Quality Assurance: software testing, issue identification, troubleshooting
- Web Development: HTML5, CSS3, JavaScript, TypeScript, React, Node.js, Express.js
- Databases and Tools: PostgreSQL, MongoDB, Prisma ORM, Git

Professional Experience

TV Engineer, Bangladesh Television (BTV), Engineering Section

Dhaka, Bangladesh | Full-time | June 2020 - July 2026

- Supported installation, commissioning and technical supervision of 10 kW analogue-to-digital television transmitters and UHF antenna systems at different sub-centers.
- Assisted project and technical teams with implementation, coordination and operations for the nationwide digital terrestrial television broadcasting project.
- Supervised installation of a 2500 kVA electrical substation at BTV Dhaka Centre.
- Participated in synchronization and commissioning of two 500 kVA generators.
- Supported installation and supervision of UPS, AVR, generators and electrical substations for broadcasting infrastructure.
- Worked in the QA team during implementation and launch of BTV's mobile television and digital service: testing, quality assurance, issue identification and coordination with technical teams.
- Contributed to modification and improvement of the BTV website, focusing on UI and user-facing interface work.
- Collaborated with technical, software and project team members on digital services, troubleshooting and implementation.

Additional Software Projects

- Vehicle Rental System: REST API using Node.js, TypeScript, Express.js and PostgreSQL, with JWT authentication and role-based authorization.
- Plant Hub and Appointment Doctor: full-stack web applications using React, Node.js, Express.js and MongoDB.

Education

- MSc, Electrical and Electronic Engineering, University of Dhaka, 2015. CGPA 3.61 / 4.00
- BSc, Applied Physics Electronics and Communication Engineering, University of Dhaka, 2013. CGPA 3.21 / 4.00

Training and Certifications

- Next Level Web Development, Programming Hero (online), Dec 2025 - Jun 2026
- Complete Web Development Course, Programming Hero (online), Dec 2024 - Jun 2025